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CHEM E 480 PROCESS DYNAMICS AND CONTROL SYSTEMS
LAB 8 REFLECTION – FEEDBACK & FEEDFORWARD CONTROL

Please briefly answer each of the following prompts in one word document, which you will submit through Canvas.

SUMMARY & REFLECTION QUESTIONS

1. What was the goal of Lab 8? Include what data you collected and what you did with that data. How did it relate to class?
   The goal of Lab 8 was to add feedforward trim to a PID controller on Q1 so the system could reject a temperature disturbance coming from Q2. We collected time, Q1, Q2, T1, T2, and the T1 set point while running the Arduino/TCLab. We first used the temperature response to estimate the disturbance effect, then used that information in the feedforward trim term. The main run stepped T1 to 40 °C, turned Q2 on as the disturbance from about 200 to 400 s, and then turned it back off. This was directly related to class because it combined feedback PID control with feedforward disturbance rejection and used the same process gains and deviation variables from our dynamic models.
2. What is feedforward trim? What equation describes feedforward trim (only, not PID control)? Define each variable.
   Feedforward trim is an extra adjustment added to the controller output based on a measured disturbance before that disturbance fully affects the controlled variable. For this lab, the feedforward trim term was:
   $G_{ff}(T_2') = K_{ff}T_2' = -(K_d/K_p)T_2' = -(K_{21}/K_{11})T_2'$
   G_{ff} is the feedforward trim contribution to the Q1 heater output. K_{ff} is the feedforward trim gain. T2′ is the deviation in the measured disturbance, meaning T2 - T2,ss. K_d, or K21, is the disturbance gain showing how the Q2/T2 disturbance affects T1. K_p, or K11, is the process gain showing how Q1 affects T1.

3. Why should the feedforward trim gain, K_{ff}, be between -1 and 0 (particularly, why should it be negative)?
   K_{ff} should be negative because the disturbance from Q2/T2 adds heat to the system, tending to raise T1. To cancel that effect, the feedforward trim needs to reduce Q1 rather than add even more heat. In other words, when T2 goes up, Q1 should be trimmed down. The value is usually between -1 and 0 because the disturbance effect is real but not stronger than the direct Q1 effect on T1. If K_{ff} is too negative, the controller can over-correct, drive Q1 too low, and cause T1 to undershoot or oscillate.

4. Why can't we use feedforward trim in isolation?
   Feedforward trim alone because it only responds to the measured disturbance. It does not know whether T1 is actually at the set point. Feedforward also depends on the model or gain being accurate, and real systems have delays, noise, heat loss, and other disturbances that are not perfectly measured. The PID feedback is still needed to correct the remaining error and bring T1 back to the set point.
5. In the "Run Arduino" section, what was the disturbance input trend qualitatively? How did this impact T2 relative to T1?
   Qualitatively, Q2 was off at first, then rose to 100% around 200 s, remained high until about 400 s, and then fell back to 0%. When Q2 turned on, T2 rose quickly and became

much hotter than T1. T1 was affected by that heat transfer, but not as strongly as T2 because the PID controller with feedforward trim reduced Q1 to help compensate. So T2 clearly showed the disturbance, while T1 remained much closer to the 40 °C set point.

6. In the “Run Arduino” section, how well did the PID controller with feedforward trim do in helping the system achieve the set point for the process?
   Overall, the PID controller with feedforward trim worked pretty well. T1 reached the 40 °C set point after the initial set point change, with some overshoot to about 41-42 °C. During the Q2 disturbance, T2 rose a lot, but T1 stayed close to 40 °C, mostly around 39.6-40.9 °C during the disturbance period. After Q2 was turned off, T1 dipped a little below the set point and then recovered. It was not perfect because Q1 was still pulsing/oscillating, but the controller did a good job rejecting most of the disturbance and keeping T1 near the target.